

# 1 Coil-based manufacturing: Configuration

## Purpose

You use this documentation, if one or several of the following conditions are true for you.

- You use the product "MPL" for the processing of rolls (coils) in your production and you want to be informed about the configuration options.
- You want to use the product "MPL" for the processing of rolls in your production.
- You want to be informed about the possibilities of specific functions that are designed for a production using rolls.

## Requirements

If you want to use the functions of the coil-based manufacturing, you must make the configurations listed in the following.

## Procedure

### Workplace and resource configuration

Configure the workplace that you want to use as follows:

- The workplace has the "workplace type" = S (cutting unit)
- The workplace is configured to use "Batch management" = L
- The workplace has a "Preceding material buffer" and a "Subsequent material buffer"
- The "Automatic generation of batch numbers for production batches (MPL)" is enabled using "J"

### Operation configuration

Configure the operation that produces the order as follows:

The operation has the option

- "Batch management requirement" = active and
- "Cutting OP" = T or M

If the operation is a mother operation, the option "Branch OP = M" is set for the operation and the field "Mother OP" includes an own MES order number.

If the operation is a child operation, the option "Branch OP = K" is set for the operation and the field "Mother OP" includes the MES order number of the respective mother operation.

The option "Cutting OP" is either set to "M – Standard numbering" or to "T – Numbering of daughter rolls".



You should only set the option "Cutting OP" = "T – Numbering of daughter rolls" after careful analysis. The batch numbers, which are created if the option "T – Numbering of daughter rolls" is set, contain the batch number of the input batch + suffix consisting of "-" and a consecutive number of 4 digits. This can have the effect that the batch numbers created are longer than the batch number length specified in the basic settings.

## Alternative quantity units

A material that is supplied on coils and is available as lengths often has more than one quantity unit. But a batch has only one quantity unit. If you want to integrate further quantity units, you can use "Activity" 1 to 6 ("Activity" = batch with further quantity unit). Therefore use "Activity" 1 to 6 to save other quantity units for a batch, if required. The "Activities" are configured via customization.

## Sample configuration

### Scenario 1

1 order with 3 operations, 1 mother operation and 2 child operations. A cut produces 3 output batches. It is planned to perform 10 output batch changes with this order. The input width of the roll is 1000 cm. The output widths of the operations are as follows:

- Mother operation: 300 cm
- Child operation 1: 200 cm
- Child operation 2: 500 cm

### Result of scenario 1

| Tab     | Field                     | Mother operation | Child operation 1 | Child operation 2 |
|---------|---------------------------|------------------|-------------------|-------------------|
| General | Order                     | 100042           | 100042            | 100042            |
| General | OP                        | 0010             | 0020              | 0030              |
| CBM     | Number of rolls           | 30               | 30                | 30                |
| CBM     | Input width               | 1000             | 1000              | 1000              |
| CBM     | Output width              | 300              | 200               | 500               |
| CBM     | Seam width                | 0                | 0                 | 0                 |
| CBM     | Surface per piece         | 300000           | 200000            | 500000            |
| CBM     | Mass per unit area        | 1250             | 1250              | 1250              |
| CBM     | Casing weight             | 200              | 200               | 200               |
| CBM     | Cutting OP                | M                | M                 | M                 |
| CBM     | Branch OP                 | M                | K                 | K                 |
| CBM     | Mother OP                 | 1000420010       | 1000420010        | 1000420010        |
| CBM     | Daughter rolls/cut        | 1                | 1                 | 1                 |
| CBM     | Daughter rolls/cut, total | 3                | 3                 | 3                 |

## Data transfer via interface

Specific segments are used to transfer the required data via interface. The segments are the following:

| Message type    | Segment          |
|-----------------|------------------|
| <b>HY72PPS</b>  | HY72_AG_RF_001_A |
| <b>ZPPORDER</b> | Z2AG_RF000X000   |